

```
Ubuntu 24.04.1 LTS X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ps
  PID TTY          TIME CMD
   735 pts/0    00:00:00 bash
  1914 pts/0    00:00:00 ps
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.1   0.3  21752 13056 ?        Ss   09:20   0:00 /sbin/init
root         2   0.0   0.0   2776  1920 ?        Sl   09:20   0:00 /init
root         8   0.0   0.0   2776    4 ?        Sl   09:21   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        44   0.0   0.4  66828 17304 ?        Ss   09:21   0:00 /usr/lib/systemd/systemd-journald
root        88   0.0   0.1  24136  6268 ?        Ss   09:21   0:00 /usr/lib/systemd/systemd-udevd
root       101   0.0   0.0 153068   640 ?        Ssl  09:21   0:00 snapfuse /var/lib/snapd/snaps/ngrok_394.snap /snap/ngrok/394 -o ro,nodev,allow_other,suid
root       104   0.0   0.0 153068   660 ?        Ssl  09:21   0:00 snapfuse /var/lib/snapd/snaps/core22_2411.snap /snap/core22/2411 -o ro,nodev,allow_other,
root       109   0.0   0.0 152936   172 ?        Ssl  09:21   0:00 snapfuse /var/lib/snapd/snaps/ngrok_390.snap /snap/ngrok/390 -o ro,nodev,allow_other,suid
root       112   0.0   0.0 227832  2284 ?        Ssl  09:21   0:00 snapfuse /var/lib/snapd/snaps/snapd_26865.snap /snap/snapd/26865 -o ro,nodev,allow_other,
systemd+   159   0.0   0.2  21460 11920 ?        Ss   09:21   0:00 /usr/lib/systemd/systemd-resolved
systemd+   160   0.0   0.1  91028  6504 ?        Ssl  09:21   0:00 /usr/lib/systemd/systemd-timesyncd
root       178   0.0   0.0   4236  2672 ?        Ss   09:21   0:00 /usr/sbin/cron -f -P
message+   179   0.0   0.1   9624  5140 ?        Ss   09:21   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
mongodb    186   0.6   5.6 3792644 225596 ?        Ssl  09:21   0:06 /usr/bin/mongod --config /etc/mongod.conf
root       192   0.0   1.1 1923940 44960 ?        Ssl  09:21   0:00 /usr/lib/snapd/snapd
root       194   0.0   0.2 17984   8424 ?        Ss   09:21   0:00 /usr/lib/systemd/systemd-logind
root       212   0.0   0.1 12024   7684 ?        Ss   09:21   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root       217   0.0   0.0   3160  1200 hvc0     Ss+  09:21   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 vt220
root       233   0.0   0.0   3116  1100 tty1     Ss+  09:21   0:00 /sbin/agetty -o -p -- \u --noclear - linux
syslog     239   0.0   0.1 222508   7172 ?        Ssl  09:21   0:00 /usr/sbin/rsyslogd -n -iNONE
root       240   0.1   1.3 1941172 55064 ?        Ssl  09:21   0:01 /usr/bin/containerd
root       246   0.0   0.5 107008 22284 ?        Ssl  09:21   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
root       342   0.0   2.1 1983072 85764 ?        Ssl  09:21   0:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
root       732   0.0   0.0   2780   204 ?        Ss   09:21   0:00 /init
root       733   0.0   0.0   2780   208 ?        S   09:21   0:00 /init
alan       735   0.0   0.2  10164  9388 pts/0    Ss   09:21   0:00 -bash
root       736   0.0   0.1   6688  4528 pts/1    Ss   09:21   0:00 /bin/login -f
alan       824   0.0   0.2  20356 11516 ?        Ss   09:21   0:00 /usr/lib/systemd/systemd --user
alan       825   0.0   0.0   21156  1724 ?        S   09:21   0:00 (sd-pam)
alan       921   0.0   0.2  10032  8936 pts/1    S+   09:21   0:00 -bash
root     1461   0.0   0.0   2780   208 ?        Ss   09:23   0:00 /init
root     1462   0.0   0.0   2780   212 ?        S   09:23   0:00 /init
alan     1470   0.0   0.2  10164  9432 pts/2    Ss   09:23   0:00 -bash
alan     1910   0.0   0.3  22252 13036 pts/2    Sl+  09:32   0:00 vi day-01-notes.yml
alan     1919   0.0   0.1   8280  4164 pts/0    R+   09:36   0:00 ps aux
```

mem

Total: 3.82 GiB
Used: 884 MiB
23%

disks

swap 2.00 GiB
Used: 0%
Free: 100% 0 Byte
2.00 GiB

io

Available: 2.96 GiB
77%

Cached: 355 MiB
9%

Free: 2.78 GiB
73%

net 172.26.80.131

sync auto zero kb eth0 n>

10K

download (0 bits)
Top: (21.3 Kibps)
Total: 134 KiB
0 Byte/s (0 bits)
Top: (0 bits)
Total: 29.6 KiB
upload

10K

proc filter

Pid	Program	Command	Threads	User	MemB	Cpu%
186	mongod	/usr/bin/mongod --config /etc/	59	mongodb	232M	0.1
342	dockerd	/usr/bin/dockerd -H fd:// --co	10	root	84M	0.0
240	containerd	/usr/bin/containerd	10	root	54M	0.0
192	snappd	/usr/lib/snappd/snappd	10	root	44M	0.0
246	unattended-upgr	/usr/bin/python3 /usr/share/un	2	root	22M	0.0
2244	bttop	bttop	2	alan	20M	0.0
44	systemd-journal	/usr/lib/systemd/systemd-journ	1	root	17M	0.0
2095	wsl-pro-service	/usr/libexec/wsl-pro-service	8	root	15M	0.0
1	systemd	/sbin/init	1	root	13M	0.0
159	systemd-resolve	/usr/lib/systemd/systemd-resol	1	systemd-r+	12M	0.0
824	systemd	/usr/lib/systemd/systemd --use	1	alan	11M	0.0
1470	bash	-bash	1	alan	9.2M	0.0
735	bash	-bash	1	alan	9.2M	0.0
921	bash	-bash	1	alan	8.7M	0.0
194	systemd-logind	/usr/lib/systemd/systemd-login	1	root	8.2M	0.0
212	sshd	sshd: /usr/sbin/sshd -D [liste	1	root	7.5M	0.0
239	rsyslogd	/usr/sbin/rsyslogd -n -iNONE	4	syslog	7.0M	0.0
160	systemd-timesyn	/usr/lib/systemd/systemd-times	2	systemd-t+	6.4M	0.0
88	systemd-udev	/usr/lib/systemd/systemd-udev	1	root	6.1M	0.0
179	dbus-daemon	@dbus-daemon --system --adres	1	messagebus	5.0M	0.0
736	login	/bin/login -f	1	root	4.4M	0.0
2077	nginx	nginx: worker process	1	www-data	4.1M	0.0
2078	nginx	nginx: worker process	1	www-data	4.1M	0.0
2079	nginx	nginx: worker process	1	www-data	4.1M	0.0
178	cron	/usr/sbin/cron -f -P	1	root	2.6M	0.0
112	snappfuse	snappfuse /var/lib/snappd/snaps/	4	root	2.2M	0.0
2	init-systemd(Ub	/init	2	root	1.9M	0.0
825	(sd-pam)	(sd-pam)	1	alan	1.7M	0.0
217	agetty	/sbin/agetty -o -p -- \u --noc	1	root	1.2M	0.0
233	agetty	/sbin/agetty -o -p -- \u --noc	1	root	1.1M	0.0
2076	nginx	nginx: master process /usr/sbi	1	root	1.0M	0.0
184	snappfuse	snappfuse /var/lib/snappd/snaps/	3	root	660K	0.0
101	snappfuse	snappfuse /var/lib/snappd/snaps/	3	root	640K	0.0
1462	Relay(1470)	/init	1	root	212K	0.0
733	Relay(735)	/init	1	root	208K	0.0
1461	SessionLeader	/init	1	root	208K	0.0
732	SessionLeader	/init	1	root	204K	0.0
109	snappfuse	snappfuse /var/lib/snappd/snaps/	3	root	172K	0.0
8	init	plan9 --control-socket 7 --log	2	root	4.0K	0.0

select

info

terminate

kill

signals

2/39

33°C
Mostly sunny

Search

ENG

IN

91%

3:28 PM
01-Jun-26

```

212
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ pgrep ssh
2095
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ pgrep vi
[1] 2338
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ sleep 500 &
[1]+  Running                  sleep 500 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[1]+  Running                  sleep 500 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ pgrep sleep
2338
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ kill 2338
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[1]+  Terminated              sleep 500
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ |

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```
Ubuntu 24.04.1 LTS X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ sleep 1000
^Z
[2]+  Stopped                  sleep 1000
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ sleep 1001 &
[3] 3269
[1]  Done                      sleep 300
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ sleep 1010 &
[4] 3270
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
  command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
  command 'jps' from deb openjdk-17-jdk-headless (17.0.18+8-1~24.04.1)
  command 'jps' from deb openjdk-21-jdk-headless (21.0.10+7-1~24.04)
  command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
  command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
  command 'jps' from deb openjdk-8-jdk-headless (8u482-ga-us1-0ubuntu1~24.04)
  command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
  command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
  command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
  command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]+  Stopped                  sleep 1000
[3]  Running                  sleep 1001 &
[4]-  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ fg %2
sleep 1000
^Z
[2]+  Stopped                  sleep 1000
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ bg %2
[2]+  sleep 1000 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
  command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
  command 'jps' from deb openjdk-17-jdk-headless (17.0.18+8-1~24.04.1)
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  command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
  command 'jps' from deb openjdk-8-jdk-headless (8u482-ga-us1-0ubuntu1~24.04)
  command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
  command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
```

33°C Mostly sunny 3:44 PM 01-Jun-26

```
Ubuntu 24.04.1 LTS  X  Windows PowerShell  X  +  v

alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
command 'jps' from deb openjdk-17-jdk-headless (17.0.18+8-1~24.04.1)
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command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
command 'jps' from deb openjdk-8-jdk-headless (8u482-ga-us1-0ubuntu1~24.04)
command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]+  Stopped                  sleep 1000
[3]   Running                  sleep 1001 &
[4]-  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ fg %2
sleep 1000
^Z
[2]+  Stopped                  sleep 1000
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ bg %2
[2]+  sleep 1000 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
command 'jps' from deb openjdk-17-jdk-headless (17.0.18+8-1~24.04.1)
command 'jps' from deb openjdk-21-jdk-headless (21.0.10+7-1~24.04)
command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
command 'jps' from deb openjdk-8-jdk-headless (8u482-ga-us1-0ubuntu1~24.04)
command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]   Running                  sleep 1000 &
[3]-  Running                  sleep 1001 &
[4]+  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ |
```

33°C
Mostly sunny



ENG
IN 91% 3:44 PM
01-Jun-26

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command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]+  Stopped                  sleep 1000
[3]   Running                  sleep 1001 &
[4]-  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ fg %2
sleep 1000
^Z
[2]+  Stopped                  sleep 1000
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ bg %2
[2]+  sleep 1000 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
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command 'jps' from deb openjdk-21-jdk-headless (21.0.10+7-1~24.04)
command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
command 'jps' from deb openjdk-8-jdk-headless (8u482-ga~us1-0ubuntu1~24.04)
command 'jps' from deb openjdk-19-jdk-headless (19.0.2+7-4)
command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
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Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]   Running                  sleep 1000 &
[3]-  Running                  sleep 1001 &
[4]+  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs -l
[2]   3268 Running              sleep 1000 &
[3]-  3269 Running              sleep 1001 &
[4]+  3270 Running              sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ps -eo pid,lstart,cmd | grep sleep
3268 Mon Jun  1 10:13:18 2026 sleep 1000
3269 Mon Jun  1 10:13:32 2026 sleep 1001
3270 Mon Jun  1 10:13:43 2026 sleep 1010
3292 Mon Jun  1 10:20:35 2026 grep --color=auto sleep
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$
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```
command 'jps' from deb openjdk-20-jdk-headless (20.0.2+9-1)
command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]+  Stopped                  sleep 1000
[3]   Running                  sleep 1001 &
[4]-  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ fg %2
sleep 1000
^Z
[2]+  Stopped                  sleep 1000
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ bg %2
[2]+  sleep 1000 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jpbs
Command 'jpbs' not found, did you mean:
command 'gpbs' from deb gnustep-back-common (0.30.0-2build1)
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command 'jps' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
command 'jps' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
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command 'jps' from deb openjdk-22-jdk-headless (22~22ea-1)
command 'apbs' from deb apbs (3.4.1-5build4)
Try: sudo apt install <deb name>
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs
[2]   Running                  sleep 1000 &
[3]-  Running                  sleep 1001 &
[4]+  Running                  sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ jobs -l
[2]   3268 Running              sleep 1000 &
[3]-  3269 Running              sleep 1001 &
[4]+  3270 Running              sleep 1010 &
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ps -eo pid,lstart,cmd | grep sleep
3268 Mon Jun  1 10:13:18 2026 sleep 1000
3269 Mon Jun  1 10:13:32 2026 sleep 1001
3270 Mon Jun  1 10:13:43 2026 sleep 1010
3292 Mon Jun  1 10:20:35 2026 grep --color=auto sleep
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$
```

```
Ubuntu 24.04.1 LTS X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ pwd
/home/alan/DevOps-Journey-Hero_to_Zero/linux_practice/day-03
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml first.py nohup.out
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls -a
. . . day-01-notes.yml.swp day-01-notes.yml first.py nohup.out
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls -la
total 32
drwxr-xr-x 2 alan alan 4096 Jun 1 10:31 .
drwxr-xr-x 3 alan alan 4096 Jun 1 09:23 ..
-rw-r--r-- 1 alan alan 12288 Jun 1 10:31 .day-01-notes.yml.swp
-rw-r--r-- 1 alan alan 1013 Jun 1 10:05 day-01-notes.yml
-rw-r--r-- 1 alan alan 25 Jun 1 10:31 first.py
-rw----- 1 alan alan 30 Jun 1 10:32 nohup.out
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ mkdir test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml first.py nohup.out test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ cd test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/test$ pwd
/home/alan/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/test$ cd ..
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ mkdir practice_commands
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml first.py nohup.out practice_commands test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ cd practice_commands/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ touch dummyfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
dummyfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cat >> dummyfile.txt
This is dummy file
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cat dummyfile.txt
This is dummy file
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
dummyfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cd ..
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ cp first.py practice_commands/dummyfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml first.py nohup.out practice_commands test
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03$ cd practice_commands/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
dummyfile.txt
```

34°C Mostly sunny 4:09 PM 01-Jun-26

```
Ubuntu 24.04.1 LTS X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
dummyfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ mv dummyfile.txt newfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
newfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls -l
total 4
-rw-r--r-- 1 alan alan 25 Jun 1 10:39 newfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
newfile.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ mkdir suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
newfile.txt suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ mkdir alan_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
alan_devops newfile.txt suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cp newfile.txt suraj_devops/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ mv newfile.txt alan_devops/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
alan_devops suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ tree
.
├── alan_devops
│   └── newfile.txt
└── suraj_devops
    └── newfile.txt

3 directories, 2 files
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ |
```

34°C Mostly sunny

Search

ENG IN 86% 4:11 PM 01-Jun-26

```
Ubuntu 24.04.1 LTS  X  Windows PowerShell  X  +  v

alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
alan_devops  suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ touch file.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
alan_devops  file.txt  suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ rm file.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ rm -r alan_devops/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ls
suraj_devops
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ rm -rf suraj_devops/
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ vi practice.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cat practice.txt
HI This is AlanDevOps
How can i help you ?
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ nano practice.txt
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ cat practice.txt
HI This is AlanDevOps
How can i help you ?
keep doing work
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ echo "hello world"
hello world
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ head -n 1 practice.txt
HI This is AlanDevOps
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ tail -1 practice.txt
keep doing work
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ tail -f practice.txt
HI This is AlanDevOps
How can i help you ?
keep doing work
This is suraj
|

23°C  Mostly sunny
Search
ENG IN 84% 4:14 PM 01-Jun-26
```

```
Ubuntu 24.04.1 LTS
alan_devops file.txt suraj_devops
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ rm file.txt
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ rm -r alan_devops/
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ ls
suraj_devops
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ rm -rf suraj_devops/
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ vi practice.txt
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat practice.txt
HI This is AlanDevOps
How can i help you ?
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ nano practice.txt
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat practice.txt
HI This is AlanDevOps
How can i help you ?
keep doing work
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ echo "hello world"
hello world
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ head -n 1 practice.txt
HI This is AlanDevOps
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ tail -1 practice.txt
keep doing work
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ tail -f practice.txt
HI This is AlanDevOps
How can i help you ?
keep doing work
This is suraj
This line is add by another terminal
What are doing now
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\SURAJ> wsl -d ubuntu-24.04
alan@Matrix:/mnt/c/Users/SURAJ$ cd /home/alan/DevOps-Journay-Hero_to_Zero/li
nux_practice/day-03/
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ vi day-01-n
otes.yml
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ ls
day-01-notes.yml first.py nohup.out practice_commands test
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03$ cd practice
_commands/
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ ls
practice.txt
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat >> practice.txt
This is suraj
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat >> this line add by another terminal
cat: line: No such file or directory
cat: add: No such file or directory
cat: by: No such file or directory
cat: another: No such file or directory
cat: terminal: No such file or directory
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat >> practice.txt
This line is add by another terminal
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_com
mands$ cat >> practice.txt
What are doing now
```

```
Windows PowerShell X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_commands$ hostname
Matrix
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_commands$ ip route
default via 172.26.80.1 dev eth0 proto kernel
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
172.18.0.0/16 dev br-15eb87f288d2 proto kernel scope link src 172.18.0.1 linkdown
172.19.0.0/16 dev br-eeaa54aaa155 proto kernel scope link src 172.19.0.1 linkdown
172.26.80.0/20 dev eth0 proto kernel scope link src 172.26.80.131
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_commands$ hostname -I
172.26.80.131 172.17.0.1 172.19.0.1 172.18.0.1 fc00:f853:ccd:e793::1
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_commands$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:33:ee:5e brd ff:ff:ff:ff:ff:ff
    inet 172.26.80.131/20 brd 172.26.95.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe33:ee5e/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 2a:20:73:d6:7a:81 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
4: br-eeaa54aaa155: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 76:8c:d2:13:32:73 brd ff:ff:ff:ff:ff:ff
    inet 172.19.0.1/16 brd 172.19.255.255 scope global br-eeaa54aaa155
        valid_lft forever preferred_lft forever
    inet6 fc00:f853:ccd:e793::1/64 scope global nodad
        valid_lft forever preferred_lft forever
5: br-15eb87f288d2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether be:6a:d7:5e:89:00 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-15eb87f288d2
        valid_lft forever preferred_lft forever
alan@Matrix:~/DevOps-Journay-Hero_to_Zero/linux_practice/day-03/practice_commands$ dig google.com
; <<>> DiG 0.18.39-Ubuntu0.24.04.5-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<- opcode: QUERY, status: NOERROR, id: 22398
```

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```
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP group default qlen 1000
   link/ether 00:15:5d:33:ee:5e brd ff:ff:ff:ff:ff:ff
   inet 172.26.80.131/20 brd 172.26.95.255 scope global eth0
       valid_lft forever preferred_lft forever
   inet6 fe80::215:5dff:fe33:ee5e/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 2a:20:73:d6:7a:81 brd ff:ff:ff:ff:ff:ff
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
4: br-eeaa54aaa155: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether 76:8c:d2:13:32:73 brd ff:ff:ff:ff:ff:ff
   inet 172.19.0.1/16 brd 172.19.255.255 scope global br-eeaa54aaa155
       valid_lft forever preferred_lft forever
   inet6 fc00:f853:ccd:e793::1/64 scope global nodad
       valid_lft forever preferred_lft forever
5: br-15eb87f288d2: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   link/ether be:6a:d7:5e:89:00 brd ff:ff:ff:ff:ff:ff
   inet 172.18.0.1/16 brd 172.18.255.255 scope global br-15eb87f288d2
       valid_lft forever preferred_lft forever
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ dig google.com

;<<>> DiG 9.18.39-ubuntu0.24.04.5-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22398
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 268     IN      A      142.250.193.46

;; Query time: 320 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Mon Jun 01 11:21:02 UTC 2026
;; MSG SIZE rcvd: 44

alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$
```

```
Windows PowerShell X Windows PowerShell X + v
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ss -tulnp
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0            127.0.0.54:53           0.0.0.0:*
udp        UNCONN     0           0            127.0.0.53%lo:53       0.0.0.0:*
udp        UNCONN     0           0            10.255.255.254:53      0.0.0.0:*
udp        UNCONN     0           0            127.0.0.1:323         0.0.0.0:*
udp        UNCONN     0           0            127.0.0.1:323         [::1]:323
tcp        LISTEN     0          4096         127.0.0.1:27017        0.0.0.0:*
tcp        LISTEN     0          4096         127.0.0.1:43049        0.0.0.0:*
tcp        LISTEN     0          1000        10.255.255.254:53      0.0.0.0:*
tcp        LISTEN     0          4096         127.0.0.54:53         0.0.0.0:*
tcp        LISTEN     0          4096         127.0.0.53%lo:53      0.0.0.0:*
tcp        LISTEN     0          4096         0.0.0.0:22            0.0.0.0:*
tcp        LISTEN     0          511         0.0.0.0:80            0.0.0.0:*
tcp        LISTEN     0          4096         [::]:22               [::]:*
tcp        LISTEN     0          511         [::]:80               [::]:*
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ ss -tulnp | grep "80"
tcp LISTEN 0      511      0.0.0.0:80      0.0.0.0:*
tcp LISTEN 0      511      [::]:80        [::]:*
alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ dig google.com
; <<>> DiG 9.18.39-0ubuntu0.24.04.5-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 18463
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 63      IN      A      142.251.43.110

;; Query time: 300 msec
;; SERVER: 10.255.255.254#53(10.255.255.254) (UDP)
;; WHEN: Mon Jun 01 11:27:03 UTC 2026
;; MSG SIZE rcvd: 44

alan@Matrix:~/DevOps-Journey-Hero_to_Zero/linux_practice/day-03/practice_commands$ |
```

Linux Command Cheat Sheet (Beginner)

1. Process management
2. File system
3. Network troubleshooting

* Process management

A process is an instance of a program currently running on a computer system. In Linux, processes are managed by the operating system's kernel, which allocates system resources and schedule processes to run on the CPU.

In Linux, processes can be categorized into two types:

1. Foreground processes: -

Foreground processes can be run interactively and wait for user input, like using an office application on Linux.

2. Background process: -

- (i) You start a task and send it to the background
- (ii) It runs silently without needing input.
- (iii) You can continue doing other work.

System vs User process

- (i) System process: - kernel starts it to manage the OS.
- (ii) User process: - user process run for user tasks.

Process management Command with description:-

- (1) `ps` : shows a list of currently running processes:-
2. `top` : Displays processes in real-time, along with CPU and memory usage.
3. `htop` : Advance version of top command with color and easier navigation.
4. `kill` : stops a process using its process ID (PID).
5. `pgrep nginx` : find the PID(s) by name.
6. `kill -9 PID` : - forcefully killed the process.
7. `killall firefox` : - kill the process by name.
8. `free -h` : - Memory usage for human readable.
9. `df -h` : - disk usage (show how much disk space is used and free on your system) for human readable (shows size like MBs).
10. `uptime` : tells the current time, how long the system has been running, how many users are logged in and how busy the CPU is.

* file system:-

The linux file system is a hierarchy of directory structures that organizes files and data on storage devices, starting from the root directory (/).

key points:-

- `/` → The root directory, the top of the hierarchy.
- `/bin` → Essential binary program (like ls, cp).
- `/home` → Users personal directories.
- `/etc` → System configuration files.
- `/var` → Variable data (logs, databases).

- /tmp → Temporary files
- /usr → User program and libraries
- /dev → Device files (hardware device)
- /lib → Essential libraries needed by system programs.

Navigation :

- (i) pwd : show the present working directory
- (ii) ls : list all the files and directories in the current folder.
- (iii) ls -l : list files and directories in the current folder with detailed information
- (iv) cd /Path/to/dir : Change directory to the specified directory
- (v) cd . : Move to one directory up (to parent folder)
- (vi)

File operations:

- (i) touch filename : create an empty file
- (ii) cp src dst : copy the file
- (iii) cp -r dir1 dir2 : Copy a directory and its contents
- (iv) mkdir foldername : create a directory
- (v) mv old new : Move or rename a file / directory
- (vi) rm filename : delete a file
- (vii) rm -rf dir : delete a directory and all its contents

View file content and edit current file

- (i) vi and nano : both command use for edit the file
- (ii) cat file : show the content of file
- (iii) echo "hello world" : display text or variables in the terminal
- (iv) head -n 10 file : show the first 10 line of file
- (v) tail -n 10 file : show the last 10 line of file
- (vi) tail -f logfile : see the latest lines of a file and watch it update in real time.

* Network Troubleshooting:-

- Network connect devices and users to exchange data.
- Can be local area network (LAN), WAN, MAN or internet.
- Network problems can cause downtime, lost productivity or revenue loss
- Troubleshooting tools help find and fix network issues:-

Connectivity:-

- (i) ping google.com : check if a device or website is reachable.
- (ii) ifconfig / ip a : shows your system's network interfaces and ip addresses.
- (iii) traceroute google.com : shows the path packets take to reach a host.

(iv) ~~net~~ mtr google.com : Combined ping and traceroute for real-time route analysis.

DNS:-

(v) nslookup example.com : Find the IP address of a domain

(vi) dig example.com : Get detailed DNS information about a domain

(vii) dig +short google.com : Quickly get just the IP address.

IP & interfaces:

(viii) ip a or ip addr : Show all network interfaces on your system along with their IP address, MAC address, and status.

(ix) ifconfig : Legacy command to view network interfaces

HTTP/HTTPS

(x) curl http://site : Make a basic HTTP request to a site.

(xi) curl -I http://site : Fetch only the HTTP headers

Download & Connectivity:

(xii) wget https://example.com/file